

Project APEX

Chapter 18 — Knowledge Graph

18.1 Purpose

The Knowledge Graph provides a structured representation of entities and relationships within Project APEX. It connects users, workflows, skills, memories, applications, tools, contexts, and decisions into a unified graph that supports reasoning and retrieval.

18.2 Objectives

- Represent workflow knowledge
- Connect related entities
- Support explainable reasoning
- Enable graph-based retrieval
- Power long-term personalization

18.3 Core Entities

- User
- Workflow
- Session
- Task
- Decision
- Skill
- Memory
- Context
- Tool
- Application
- Document

18.4 Relationship Types

- PERFORMS
- CONTAINS
- DEPENDS_ON
- USES
- GENERATED
- REFERENCES
- PREFERS
- PRECEDES

18.5 Graph Construction

Observation events are transformed into workflow entities. Context enrichment and memory encoding create relationships that are incrementally added to the Knowledge Graph. Continuous feedback updates both nodes and edges over time.

18.6 Graph Queries

The graph supports semantic navigation, dependency analysis, workflow reconstruction, recommendation generation, impact analysis, and retrieval for the RAG pipeline.

18.7 Design Principles

- Schema-driven modeling
- Explainable relationships
- Incremental updates
- Version-controlled evolution
- Graph database compatibility

18.8 Challenges

Maintaining graph consistency, preventing duplicate entities, scaling to millions of relationships, and preserving temporal history are key engineering challenges.

18.9 Role in APEX

The Knowledge Graph is the central semantic layer connecting the Observation Engine, Memory Layer, RAG pipeline, Behavior Learning Engine, LLM Layer, and Planning Engine.

Entity	Primary Role
User	Workflow owner
Workflow	High-level process
Task	Logical work unit
Decision	Choice representation
Skill	Reusable capability
Memory	Historical knowledge
Context	Situational metadata
Tool	Execution target

18.10 Chapter Summary

The Knowledge Graph serves as the semantic backbone of Project APEX by organizing entities and their relationships into a unified, explainable knowledge model. It enables efficient retrieval, reasoning, personalization, and workflow intelligence across the platform.